



PROJECT PROPOSAL.

TIBA HOSPITAL MANAGEMENT SYSTEM

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22/05522

BIT 04105: FINAL YEAR PROJECT I

BSc. INFORMATION TECHNOLOGY

1. INTRODUCTION

Hospital Management System (HMS) or Hospital Information System (HIS) is a combination of information systems which are to control all the operations of the hospital i.e. medical, administrative, financial and legal and the corresponding processing of the services. The main purpose of an HMS is focused on workflow optimization, operational effectiveness, patient care and the creation of a highly efficient data storage to support the decision-making process.

2. BACKGROUND

The healthcare sector is rapidly changing following the necessity to be efficient, transparent, and patient-centered. The quality of medical service that hospitals offer is no longer the only parameter that benchmarks hospitals, but also the effectiveness with which the hospitals utilize their resources, optimize their operations, and improve the overall experience of the patient.

In spite of these changing needs, most healthcare organizations continue to operate manual and disjointed systems, to handle such important operations as patient records, billing, scheduling, and inventory.

Such outmoded methods tend to lead to redundancy of work and inconsistency of data, and paucity of interdepartmental visibility. As a result, hospitals have to incur a greater amount of administrative expenses, there is a greater risk of errors, and delays which adversely influence the productivity of the staff and the satisfaction of the patients.

The urgency to have an all-inclusive, unified system of hospital management has never been higher. Such a system will be able to combine the administrative, clinical, and financial processes to one platform that will cause proper data flow, minimize redundancies, enhance decision-making, and eventually allow hospitals to devote increased attention to providing quality patient care.

3. PROBLEM STATEMENT

Hospitals in the modern world are working in very complicated conditions, which demand effective coordination between departments and functions. Nevertheless, the use of the fragmented systems and manual processes remains a problem. Most of the information of vital patients is usually distributed among the various platforms making it hard to access and share in real time.

Administrative processes, including billing, scheduling, and reporting, are likely to be delayed, and financial processes are not as transparent as possible to make effective decisions. Such inefficiencies not only raise the operation costs, but also undermine the quality of care provided to the patients.

Medical professionals have no choice but to devote precious hours to the administration process rather than attend to the demands of the patients, which frustrates both the medics and the patients. In the absence of a cohesive and trustworthy system of management, the hospitals will face decreased performance, ineffective use of resources, and a general decrease in patient satisfaction.

4. OBJECTIVES

The primary objective is to create the ‘Tiba hospital management system’ that will streamline the appointment process and improve access to specialized doctors ultimately delivering exceptional patient experience and superior value. The project aims to achieve the following specific objectives:

- To design and develop a responsive user-friendly interface for ease of use for both medical staff and administrative personnel.
- To design and develop a centralized database for storing and managing patient information, medical records, staff details and hospital resources.
- To develop functional modules for key hospital operations such as patient registration, appointment scheduling and billing.
- To test and evaluate the system’s performance, usability and reliability in comparison to existing manual hospital management workflows.

- To generate analytical reports on hospital performance, including patient inflow, revenue, doctor efficiency and common treatments.
- To deploy the finalized Tiba HMS on a local server and document its installation, configuration and user guide for end-users.

5. LITERATURE REVIEW

Development and History of Hospital Management Systems (HMS).

The development of Hospital Management Systems (HMS) is a positive shift in the digitalization of the healthcare sector. The earliest HMS came into existence in the 1960s and 1970s as crude mainframes that were mainly used to handle administrative functions including billing and payrolls. To a great extent, these systems were siloed and did not provide much functionality. By the 1990s with the introduction of client-server architectures, specialized modules of clinical departments such as Laboratory Information Systems (LIS) and Radiology Information Systems (RIS) were developed. In the modern world, there is the balance between the enterprise-wide solutions and Electronic Health Records (EHRs), forming seamless web-based platforms that increase the accessibility of data and coordination of care across departments.

Basic Functional Elements and Building Blocks.

The modern HMS are normally modular in their nature and they are a combination of some interdependent parts. The main modules consist of patient management, clinical (containing EHRs and Computerized Physician Order Entry [CPOE] systems), LIS/RIS, billing and financial management, inventory and supply chain management and human resource (HR) management. All these modules together tend to automate administrative functions, increase clinical accuracy, and promote workflow efficiency. In the case of CPOE systems, the latter have been found to have a tremendous impact on the decrease of medication errors.

Recorded Positive and negative effects.

HMS implementation has been linked to significant operational, clinical as well as financial gains. HMS make routine operations faster, less paperwork and redundant data entry operationally makes HMS more efficient and less costly in terms of administration. Integrated EHRs and Clinical Decision Support Systems (CDSS) are clinical where they aid in making informed decisions by providing all patient information and alerts on the likelihood of drug interactions. Automated billing and coding reduce claim rejection and speed up reimbursement process financially. In addition, HMS facilitate the use of data in decision-making because they can become central repositories of institutional analytics, where hospitals can track key performance indicators (KPIs) and optimize resource utilization.

Difficulties and Obstacles to Implementation.

Regardless of the well-established advantages, HMS implementation is complicated and consumes many resources. Software, hardware, and training that require high capital investment is a major obstacle that is also costly especially to smaller healthcare centers. Also, the resistance of users, which can be due to poor training or disruption of the work processes, is a common barrier to adoption. The problem of interoperability still exists because most systems do not have standardized protocols through which they can easily share data in healthcare institutions. Lastly, the issue of data security and privacy is becoming more urgent and HMS are the ideal targets of cyberattacks and are bound to adhere to the strict data protection policies like HIPAA and GDPR.

Existing HMS systems.

Afyapro: This is a commercial electronic health management that has been piloted in several Kenyan hospitals.

- **Strengths:** Comprehensive integration of hospital functions, supports both offline and online operations, and offers multilingual interfaces.
- **Weaknesses:** High initial setup costs, limited scalability for smaller facilities, and dependency on vendor support for customization and troubleshooting.

OpenMRS/AfyaEHMS: the AfyaEHMS is an adaptation of the open-source OpenMRS platform developed for hospitals in Machakos county to improve patient data management and streamline administrative functions.

- **Strengths:** Open-source and cost-effective, customizable to local workflows, and compatible with national health reporting systems.
- **Weaknesses:** Required significant user training and technical support, experienced challenges with workflow alignment, and faced resistance from staff used to manual system.

New Trends and Future Projections.

There have been recent developments that are reshaping HMS. The use of cloud-based solutions is emerging because it is easy to scale, economical, and the infrastructure needs lower costs. Interoperability standardization like Fast Healthcare Interoperability Resources (FHIR) is improving the sharing of data between a wide range of systems. In addition, the use of Artificial Intelligence (AI) and big data analytics makes it possible to predict patient readmission risk, optimize workforce, and support with diagnostics. Parallel to this, patient portals are taking the center stage in the design of HMS, and they encourage patient involvement and healthcare management transparency.

Conclusion

According to the literature, there is a consistent theme that HMS have shifted to be optional tools to crucial infrastructure to provide modern healthcare. Their implementation will result in higher levels of operational efficiency, clinical outcomes, and financial sustainability. However, the implementation requires proper planning, efficient change management, interoperability, and strong cybersecurity to be successful. The interest of future studies must also be to determine the ROI of high-technology HMS capabilities, specifically AI integration, in the long-term, and determine evidence-based methods to resolve the endemic implementation barriers.

6. SIGNIFICANCE OF THE STUDY

This study shall be of value to the different stakeholders within the healthcare ecosystem. These include:

- In the instance of Hospitals: It is a guide to streamline the operations into efficient, financially transparent state and service delivery in general, by incorporating an overall system.
- To Healthcare Providers It will contribute to reducing the administrative load, which will allow physicians, nurses, and other personnel to dedicate more time to working with the patient, which will improve the final results of the work.
- In regard to Patients: The study reveals how the enhancement of records benefits eventually reduces wait times and straightforward payments to enhance the patient experience and build trust in the healthcare system.
- To Policy makers and Regulators: The study can be applied to advise the policy on the digital transformation of the healthcare that can help standardize and modernize the functioning of the hospitals.
- To Researchers and Academia: The article will be informative of the current knowledge in the subject of healthcare IT adoption, the prospect in the way to develop digital health and hospital administration systems will be grounded on it.

Lastly, the significance of technology as a crucial issue in healthcare management and optimal patient-centered care is also brought out in the paper.

7. METHODOLOGY

The Agile Methodology approach of the web based 'Tiba' hospital management system will be based on the concept of incremental build and continuous feedback during the system development phase.

- **Planning and Requirement Analysis:** Speaking to hospital administrators, medical staff and other stakeholders continually to allow for a classification of needs into multiple, easily digestible chunks known as sprints.
- **System Design and Development:** Dividing the design and development of the system into different sprints that are smaller cycles where the features, such as patient management, inventory, integration with payment solutions, and appointment tracking, can be developed, tested, and improved. The Tiba HMS will be developed using modern technologies such as PHP and MySQL for backend development, HTML, CSS and JavaScript for the user interface.
- **Testing and Quality Assurance:** Integrative testing will be done constantly in every sprint with the intention of generating quick feedback on the functionality and security aspects that can be altered as soon as possible.
- **Deployment, training and Feedback:** Introducing the features step by step in phases to minimize disruptions to hospital operations with test versions that can be used for identifying users' reactions to offered options. Provide comprehensive training sessions for medical staff, administrators and IT personnel to ensure smooth adoption.
- **Continuous Monitoring and Updates:** Providing a feedback system that contains iteration and update activities that will help improve the system frequently to meet the needs of the hospital.

This design pattern will allow the project to be as adaptive as possible in terms of the development process, delivering features in incremental cycles and enhancing these cycles at any given time throughout the progress of the project.

To make sure the new Hospital Management System truly meets the needs of healthcare professionals, a thoughtful research approach will be used. The goal is to understand the real challenges hospitals face today, learn from existing systems, and uncover proven strategies that can shape a better, more effective solution. This will be achieved by:

- **Research design:** the study will adopt a descriptive and exploratory research design where descriptive elements will be used to capture the current processes

and challenges in hospital management and the exploratory aspects will focus on identifying opportunities for improvement and innovative solutions.

- Data collection methods: structured interviews and questionnaires will be administered to hospital administrators, doctors, nurses and support staff to capture their perspectives on their current systems and challenges.
- Sampling technique: a purposive sampling approach will be applied targeting respondents like the clinical staff, administrative personnel, IT staff and the patients who directly interact with hospital management systems.
- Data analysis: the qualitative data from interviews and the quantitative data from the questionnaires will be analyzed statistically to highlight the recurring challenges, needs, the trends in user satisfaction, error rates and operational delays.
- Ethical considerations: confidentiality and anonymity will be assured to all the participants to encourage honest feedback.

8. PROJECT SCHEDULING

Task No.	Task	Duration	Expected Start date	Actual Start date	Expected End date	Actual end date	Deliverable
1.	Ideation	2 days	24/09/2025	25/09/2025	26/09/2025	27/09/2025	Project idea approved
2.	Project Proposal Design	5 days	06/10/2025	8/10/2025	11/10/2025	13/10/2025	Project proposal
3	Proposal Presentation	1 day		14/10/2025		14/10/2025	Presented Proposal

4	System Requirements specification design	1 week		16/10/2025		23/10/2025	SRS Document
5	System Design Specification Document	1 week		24/10/2025		31/10/2025	SDS Document
6	Test Plan Document	1 week		04/11/2025		11/11/2025	Test Plan Document
7	System Coding and Compiling	2 weeks		14/11/2025		28/11/2025	System Development
8	System Testing	3 days		09/01/2026		13/01/2026	Test Results
9	Implementation Strategy Document Design	1 week		14/01/2026		21/01/2026	Implementation Strategy Document
10	User Manual Design	2 days		23/01/2026		26/01/2026	User Manual
11	Compilation of the system and documentation	1 week		28/01/2026		05/02/2026	Final Project Document and Compilation
12	Project Presentation	1 week					Project Presentation

9. BUDGET

ITEM	QUANTITY	COST	AVAILABILITY	USE
Laptop	1	Kshs 45,000	Unavailable	It will help in the development of the system.
Wi-Fi		Kshs 3000	Unavailable	Enable connection to the internet
Anti-virus			Available	Protection from viruses
Printing and Binding documents	One ream	Kshs 600		For publication of the documents.
Miscellaneous Expense		Kshs 2000		Other expenses

10. REFERENCES

Nishanthan, T., & Fernando, D. (2023). The Hospital Management System. ResearchGate.
https://www.researchgate.net/publication/367460409_The_Hospital_Management_System

Osman, H., & Samuel, S. (2023). Assessing the Use of Hospital Information Systems (HIS) to Support Better Delivery of Health Care. Journal of Health Informatics, 18(2), 45–58.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9322972>

Sangeetha, K., & Kumar, A. (2024). Smart Hospital Management System: Streamlining Healthcare Operations. SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4830550

Patel, V., & Singh, P. (2024). The Role and Impact of New Technologies on Healthcare. *Health Technology Review*, 8(2), 121–139. <https://link.springer.com/article/10.1007/s44250-024-00163-w>

Alanzi, T., & Alturki, R. (2024). Assessing the Effectiveness of Healthcare Management Systems in Quality Enhancement. *International Journal of Healthcare Information Systems*, 12(3), 1–12. <https://www.researchgate.net/publication/388795846>

Tuhin, A. I. (2023). *Hospital Management System*. Independent University, Bangladesh.